

Requirement Analysis Document

Online Book Store System

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Submitted to: 10Pearls Shine Internship Program

Date: 4th Aug,2026

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1. Introduction

This document presents a structured requirement analysis for an Online Book Store system. It covers the understanding of business goals and user expectations, functional and non-functional requirements, user stories, acceptance criteria written in Gherkin syntax, and a sprint-level breakdown of the work. The purpose of this document is to translate business needs into clear, testable requirements that can guide both development and QA activities.

2. Requirement Understanding

2.1 Business Goal

The primary objective of the Online Book Store is to provide customers with a secure and user-friendly platform for discovering and purchasing books online. The platform should support the complete purchasing journey, from account creation and product discovery to checkout and order confirmation.

2.2 Expected User Behavior

A typical user browses or searches for a book, reviews its details, adds it to a cart, and checks out with shipping and payment information. Registered users expect to manage their profile and view order history.

2.3 Key Workflows

1. Account Workflow: Visitor registers → logs in → manages profile.
2. Discovery Workflow: User browses catalog or searches by title/author/category → views book details.
3. Purchase Workflow: User adds books to cart → updates cart → checkout → payment → confirmation.
4. Post-Purchase Workflow: User receives confirmation → tracks order → views order history.

2.4 Assumptions

- Users must be logged in to add items to cart. Guest checkout is not assumed in this phase.
- A single currency (e.g. PKR) and single default language (English) are assumed for this phase.
- Payment processing is handled through one integrated third-party gateway (e.g. EasyPaiza/JazzCash-style), not built in-house — raw card data is never stored by the platform.
- Book inventory/stock levels are maintained by a separate catalog-management process; this system only reads and validates against them.
- Email delivery (order confirmations) is handled via a transactional email service assumed to be reliable and outside this document's control.

2.5 Out of Scope

- Guest checkout without account creation.
- Multi-currency pricing and international shipping/tax calculation.
- Book reviews, ratings, and recommendation engine.
- Admin/inventory management back-office panel.
- Native mobile applications (only responsive web is covered).

3. Functional Requirements

The following functional requirements describe the core features and system behavior expected from the Online Book Store System.

ID	Requirement	Description	Priority
FR-01	User Registration	The system shall allow new users to create an account using name, email, and password.	High
FR-02	User Login & Authentication	The system shall allow registered users to log in securely and log out of their account.	High
FR-03	Browse Book Catalog	The system shall allow users to browse the full list of available books, organized by category.	High
FR-04	Search Books	The system shall allow users to search for books by title, author, or category.	High
FR-05	View Book Details	The system shall display title, author, price, description, availability, and cover image for a selected book.	High
FR-06	Add to Cart	The system shall allow logged-in users to add in-stock books to a shopping cart.	High
FR-07	Update / Remove Cart Items	The system shall allow users to update quantity or remove items from the cart.	Medium
FR-08	Checkout & Payment	The system shall allow users to enter shipping details and complete payment via an integrated payment gateway.	High
FR-09	Order Confirmation	The system shall display an order confirmation and send a confirmation email upon successful payment.	Medium
FR-10	Manage User Account	The system shall allow users to view/update profile, address, and order history.	Medium
FR-11	Stock / Inventory Validation	The system shall validate book stock availability at the point of add-to-cart and again at checkout, blocking purchase of unavailable items.	High
FR-12	Input Validation	The system shall validate mandatory fields and formats (e.g. email format, password strength, required shipping fields) and reject submissions that fail validation with a clear error.	High

4. Non-Functional Requirements

The following non-functional requirements define the quality attributes and performance expectations of the system.

ID	Category	Requirement	Standard / Metric
NFR-01	Performance	Catalog browsing, search, and page loads shall be fast and responsive under normal load.	Search results returned within 2 seconds for 95% of requests.
NFR-02	Security	User credentials and payment data shall be protected against unauthorized access and common web attack vectors.	Passwords hashed (e.g. bcrypt); TLS 1.2+ everywhere; payment handled via a PCI-DSS compliant gateway; auth flows aligned to OWASP Top 10 guidance.

NFR-03	Availability	The platform shall remain accessible with minimal unplanned downtime.	99.5% uptime per month, monitored.
NFR-04	Usability	The interface shall be intuitive enough for first-time users to complete a purchase without assistance.	A new user completes checkout in under 5 minutes with zero support contact, per usability testing.
NFR-05	Scalability	The system shall support growth in users and catalog size without performance degradation.	Supports 10,000 concurrent users without response-time degradation beyond NFR-01 targets.
NFR-06	Accessibility	The platform shall be usable by people with disabilities.	Complies with WCAG 2.1 Level AA.
NFR-07	Compatibility	The platform shall function correctly across major browsers and device sizes.	Chrome, Firefox, Safari, Edge (latest 2 versions); responsive from 360px to desktop widths.
NFR-08	Reliability & Error Handling	The system shall fail gracefully, without data loss or duplicate transactions, and surface clear error messages.	No duplicate orders on payment retry; failed transactions leave cart state intact; all errors are user-readable, not raw system errors.

5. User Stories

The following user stories capture the system requirements from the end user's perspective.

ID	User Story	Priority
US-01	As a new visitor, I want to create an account, so that I can save my details and track my orders.	High
US-02	As a registered user, I want to log in to my account, so that I can access my cart, orders, and profile.	High
US-03	As a shopper, I want to browse books by category, so that I can discover titles that interest me.	High
US-04	As a shopper, I want to search for a book by title or author, so that I can quickly find a specific book.	High
US-05	As a shopper, I want to view full details of a book, so that I can decide whether to purchase it.	High
US-06	As a shopper, I want to add a book to my cart, so that I can purchase it later or with other books.	High
US-07	As a shopper, I want to update or remove items in my cart, so that my order matches what I intend to buy.	Medium
US-08	As a buyer, I want to check out and pay online, so that I can complete my purchase securely.	High
US-09	As a buyer, I want to receive an order confirmation, so that I have proof of purchase and delivery expectations.	Medium
US-10	As a returning user, I want to view my past orders, so that I can track deliveries or reorder books.	Low
US-11	As a shopper, I want to clearly see when a book is out of stock, so that I don't attempt to buy something unavailable.	Medium
US-12	As a buyer, I want to be notified immediately if my payment fails, so that I can retry or use a different method without losing my cart.	High

6. Requirement Traceability Matrix

The Requirement Traceability Matrix (RTM) establishes the relationship between user stories, functional requirements, and corresponding test scenarios. It ensures that each requirement is covered by one or more test cases and that all user stories can be verified during testing.

User Story	Functional Requirement(s)	Test Scenario(s)
US-01	FR-01, FR-12	SC-01, SC-02, SC-03, SC-04, SC-05
US-02	FR-02	SC-06, SC-07, SC-08, SC-09
US-03	FR-03	SC-10–SC-13 (search family)
US-04	FR-04	SC-10, SC-11, SC-12, SC-13
US-05	FR-05, FR-11	SC-14, SC-15
US-06	FR-06, FR-11	SC-16, SC-17, SC-18, SC-21
US-07	FR-07	SC-19, SC-20
US-08	FR-08, FR-12	SC-22, SC-23, SC-24, SC-25, SC-26
US-09	FR-09	SC-27, SC-28
US-10	FR-10	Covered under account management — regression only
US-11	FR-11	SC-15, SC-17, SC-26
US-12	FR-08, NFR-08	SC-24, SC-28

7. Risk-Based Test Focus

The following table identifies the key functional areas of the Online Book Store System that require prioritized testing based on their potential business impact, technical complexity, and likelihood of defects.

Functional Area	Risk Description	Business Impact	Priority	Mitigation / Test Focus
Payment & Checkout	Payment failures, duplicate transactions, incorrect order creation, or transaction timeouts may result in financial loss and poor user experience.	High	High	Perform extensive positive, negative, edge-case, and regression testing. Validate payment retries, timeout handling, duplicate transaction prevention, and successful order creation.
Authentication & Session Management	Invalid authentication or improper session handling may expose user accounts to unauthorized access.	High	High	Verify login validation, password policies, account lockout, session timeout, logout functionality, and protection against unauthorized access.
Inventory Consistency	Concurrent purchases of limited stock may result in overselling or inconsistent inventory records.	High	High	Test stock validation during Add to Cart and Checkout, including concurrent purchase scenarios and inventory synchronization.

Third-Party Payment Gateway Integration	External payment gateway failures or communication issues may interrupt the checkout process.	Medium	Medium	Validate error handling for declined payments, gateway unavailability, network interruptions, and retry mechanisms using mocked or simulated responses.
Search & Catalog Browsing	Inefficient search functionality or incorrect filtering may reduce usability and product discoverability.	Medium	Low	Verify search accuracy, filtering, sorting, empty-result handling, and response time under normal operating conditions.

8. Acceptance Criteria (Gherkin Syntax)

The following acceptance criteria define the expected system behavior for each feature using Gherkin syntax. Test scenarios include positive, negative, and edge cases to validate both functional behavior and error handling.

8.1 Feature: User Registration

Scenario SC-01 — Successful registration with valid details [Positive]

Given the user is on the registration page

When the user enters a valid name, a unique email address, and a valid password, then submits the form

Then the system should create a new account and redirect the user to the login page or homepage

Scenario SC-02 — Registration with an already-registered email [Negative]

Given the user is on the registration page and the entered email already exists in the system

When the user submits the registration form

Then the system should reject the submission and display a message indicating the email is already in use, without creating a duplicate account

Scenario SC-03 — Registration with invalid email format [Negative]

Given the user is on the registration page

When the user enters an email address that is not in a valid format (e.g. missing '@') and submits

Then the system should block submission and display a field-level validation error

Scenario SC-04 — Registration with a weak password [Negative]

Given the user is on the registration page and password policy requires a minimum length and complexity

When the user enters a password that does not meet the policy and submits

Then the system should reject the submission and clearly state the password requirements

Scenario SC-05 — Registration with a mandatory field left blank [Edge]

Given the user is on the registration page

When the user leaves a required field (e.g. name) empty and submits

Then the system should prevent submission and highlight the missing required field

8.2 Feature: User Login

Scenario SC-06 — Successful login with valid credentials [Positive]

Given the user is on the login page and has a registered, active account

When the user enters the correct email and password and submits

Then the system should authenticate the user and redirect them to their account dashboard

Scenario SC-07 — Login with an incorrect password [Negative]

Given the user is on the login page and has a registered account

When the user enters the correct email but an incorrect password

Then the system should deny access and display a generic “invalid credentials” message that does not reveal whether the email or password was wrong

Scenario SC-08 — Login with an unregistered email [Negative]

Given the user is on the login page

When the user enters an email address that has no associated account

Then the system should deny access and display the same generic “invalid credentials” message as SC-07, to avoid confirming which emails are registered

Scenario SC-09 — Account lockout after repeated failed login attempts [Edge]

Given the user has failed to log in with incorrect credentials multiple times consecutively (e.g. 5 attempts)

When the user attempts to log in again

Then the system should temporarily lock the account and inform the user, rather than allowing unlimited attempts

8.3 Feature: Search & Catalog Browsing

Scenario SC-10 — Search for a book by title [Positive]

Given the user is on the bookstore homepage

When the user enters a valid, existing book title in the search bar and submits

Then the system should display a list of relevant matching results

Scenario SC-11 — Search for a book by author name [Positive]

Given the user is on the bookstore homepage

When the user searches using a valid author name

Then the system should display all books by that author

Scenario SC-12 — Search with no matching results [Negative]

Given the user is on the bookstore homepage

When the user searches for a title/author that does not exist in the catalog

Then the system should display a clear “no results found” message rather than an empty or broken page

Scenario SC-13 — Search with an empty query [Edge]

Given the user is on the bookstore homepage

When the user submits the search with no text entered

Then the system should either show a validation prompt or return the full catalog, rather than erroring out

8.4 Feature: View Book Details

Scenario SC-14 — View details of an in-stock book [Positive]

Given the user has selected a book that is currently in stock

When the book details page loads

Then the system should display title, author, price, description, availability status, and an active ‘Add to Cart’ button

Scenario SC-15 — View details of an out-of-stock book [Edge]

Given the user has selected a book that is currently out of stock

When the book details page loads

Then the system should clearly display an ‘Out of Stock’ status and disable the ‘Add to Cart’ action

8.5 Feature: Shopping Cart

Scenario SC-16 — Add an in-stock book to the cart [Positive]

Given the user is viewing the details of a book that is in stock

When the user clicks ‘Add to Cart’

Then the system should add the book to the cart and update the cart item count

And the system should display a confirmation message to the user

Scenario SC-17 — Attempt to add an out-of-stock book to the cart [Negative]

Given a book's availability changed to out-of-stock after the page was loaded (e.g. via a direct/stale link)

When the user attempts to add it to the cart

Then the system should block the action and display a message that the item is no longer available

Scenario SC-18 — Add the same book to the cart twice [Edge]

Given the user already has one unit of a book in the cart

When the user adds the same book again from the details page

Then the system should increment the quantity for that book rather than creating a duplicate line item

Scenario SC-19 — Update quantity of an item in the cart [Positive]

Given the user has at least one book in the cart

When the user changes the quantity for that item to a valid number within available stock

Then the system should update the line total and cart total accordingly

Scenario SC-20 — Remove an item from the cart [Positive]

Given the user has at least one book in the cart

When the user clicks ‘Remove’ on a cart item

Then the system should remove the item and recalculate the cart total, updating the cart to empty state if it was the last item

Scenario SC-21 — Cart persists across logout and login [Edge]

Given the user has items in their cart and logs out without checking out

When the user logs back in later

Then the system should restore the previously saved cart contents

8.6 Feature: Checkout & Payment

Scenario SC-22 — Successful checkout with valid payment details [Positive]

Given the user has at least one in-stock book in the cart and is on the checkout page

When the user enters valid shipping details and valid payment information, then confirms the order

Then the system should process payment, create the order, and display an order confirmation with a unique order ID

Scenario SC-23 — Checkout attempt with an empty cart [Negative]

Given the user's cart is empty

When the user navigates directly to the checkout page

Then the system should prevent checkout and redirect with a message indicating the cart is empty

Scenario SC-24 — Checkout with a declined payment [Negative]

Given the user has a valid cart and enters payment details that the gateway declines

When the user submits the payment

Then the system should not create an order, should keep the cart intact, and should display a clear payment-failure message allowing the user to retry

Scenario SC-25 — Checkout with incomplete shipping details [Negative]

Given the user is on the checkout page with items in the cart

When the user submits the order with a required shipping field left blank

Then the system should block submission and highlight the missing field, without charging payment

Scenario SC-26 — Stock becomes unavailable between add-to-cart and checkout [Edge]

Given a book in the user's cart sells out (stock reaches zero) after it was added but before checkout is completed

When the user proceeds to checkout

Then the system should detect the stock conflict, remove or flag the affected item, and prevent the user from completing payment for it

8.7 Feature: Order Confirmation

Scenario SC-27 — Order confirmation displayed and emailed after successful payment [Positive]

Given the user has just completed a successful payment

When the order is created

Then the system should display an on-screen confirmation with order summary and order ID

And the system should send a confirmation email to the user's registered address

Scenario SC-28 — No order is created when payment fails [Negative]

Given the user's payment attempt fails at checkout

When the failure is returned by the payment gateway

Then the system should not generate an order confirmation, not send a confirmation email, and return the user to checkout with the cart preserved

9. Sprint Planning

User stories are organized into development sprints based on feature dependencies and implementation priority.

Sprint	Focus Area	Included User Stories
Sprint 1	Account & Access	US-01 Registration, US-02 Login/Authentication
Sprint 2	Catalog & Discovery	US-03 Browse Catalog, US-04 Search Books, US-05 View Details, US-11 Out-of-Stock Visibility
Sprint 3	Cart & Checkout	US-06 Add to Cart, US-07 Update/Remove Cart, US-08 Checkout & Payment, US-12 Payment Failure Handling
Sprint 4	Post-Purchase	US-09 Order Confirmation, US-10 Order History

10. Conclusion

This requirement analysis document identifies the functional and non-functional requirements of the Online Book Store system and translates business needs into testable specifications. The documented user stories, acceptance criteria, traceability matrix, and sprint plan provide a foundation for development, manual testing, and future regression testing activities.